A Additional Oralytics RL Algorithm Facts

A.1 Algorithm State Space

$S_{i,t} \in \mathbb{R}^d$ represents the ith participant's state at decision point t, where d is the number of variables describing the participant's state.

Baseline and Advantage State Features Let $f(S_{i,t}) \in \mathbb{R}^5$ denote the features used in the algorithm's model for both the baseline reward function and the advantage.

These features are:

1. Time of Day (Morning/Evening) $\in \{0, 1\}$
2. $\bar{B}$: Exponential Average of OSCB Over Past 7 Days (Normalized) $\in [-1, 1]$
3. $\bar{A}$: Exponential Average of Engagement Prompts Sent Over Past 7 Days (Normalized) $\in [-1, 1]$
4. Prior Day App Engagement $\in \{0, 1\}$
5. Intercept Term $= 1$

Feature 1 is 0 for morning and 1 for evening. Features 2 and 3 are $\bar{B}_{i,t} = c_\gamma \sum_{j=1}^{14} \gamma^{j-1} Q_{i,t-j}$ and $\bar{A}_{i,t} = c_\gamma \sum_{j=1}^{14} \gamma^{j-1} A_{i,t-j}$ respectively, where $\gamma = 13/14$ and $c_\gamma = \frac{1-\gamma}{1-\gamma^{14}}$. Recall that $Q_{i,t}$ is the proximal outcome of OSCB and $A_{i,t}$ is the treatment indicator. Feature 4 is 1 if the participant has opened the app in focus (i.e., not in the background) the prior day and 0 otherwise. Feature 5 is always 1. For full details on the design of the state space, see Section 2.7 in Trella et al. (2024a).

A.2 Reward Model

The reward model (i.e., model of the mean reward given state s and action a) used in the Oralytics trial is a Bayesian linear regression model with action centering (Liao et al. 2019):

$$r_\theta(s, a) = f(s)^T \alpha_0 + \pi f(s)^T \alpha_1 + (a - \pi) f(s)^T \beta + \epsilon \tag{5}$$

where $\theta = [\alpha_0, \alpha_1, \beta]$ are model parameters, π is the probability that the RL algorithm selects action $a = 1$ in state s and $\epsilon \sim \mathcal{N}(0, \sigma^2)$. We call the term $f(S_{i,t})^T \beta$ the advantage (i.e., advantage of selecting action 1 over action 0) and $f(S_{i,t})^T \alpha_0 + \pi_{i,t} f(S_{i,t})^T \alpha_1$ the baseline. The priors are $\alpha_0 \sim \mathcal{N}(\mu_{\alpha_0}, \Sigma_{\alpha_0})$, $\alpha_1 \sim \mathcal{N}(\mu_\beta, \Sigma_\beta)$, $\beta \sim \mathcal{N}(\mu_\beta, \Sigma_\beta)$. Prior values for $\mu_{\alpha_0}, \Sigma_{\alpha_0}, \mu_\beta, \Sigma_\beta, \sigma^2$ are specified in Section A.3. For full details on the design of the reward model, see Section 2.6 in Trella et al. (2024a).

A.3 Prior

Table 5 shows the prior distribution values used by the RL algorithm in the Oralytics trial. For full details on how the prior was constructed, see Section 2.8 in Trella et al. (2024a).

Parameter	Oralytics Pilot
σ^2: noise variance	3878
μ_{α_0}: prior mean of the baseline state features	$[18, 0, 30, 0, 73]^T$
Σ_{α_0}: prior variance of the baseline state features	$\text{diag}(73^2, 25^2, 95^2, 27^2, 83^2)$
μ_β: prior mean of the advantage state features	$[0, 0, 0, 53, 0]^T$
Σ_β: prior variance of the advantage state features	$\text{diag}(12^2, 33^2, 35^2, 56^2, 17^2)$

Table 5: **Prior Used in Oralytics Trial.** Values are rounded to the nearest integer. Recall that the ordering of the features is the same as described in Section A.1: Time of Day, Exponential Average of Brushing Over Past 7 Days (Normalized), Exponential Average of Engagement Prompts Sent Over Past 7 Days (Normalized), Prior Day App Engagement, Intercept Term.

B Simulation Environment

We created a simulation environment using the Oralytics trial data in order to replicate the trial under different true environments. Although the trial ran with 79 participants, due to an engineering issue, data for 7 out of the 79 participants was incorrectly saved and thus their data is unviable. Therefore, the simulation environment is built off of data from the 72 unaffected participants. Replications of the trial are useful to (1) re-evaluate design decisions that were made and (2) have a mechanism for resampling to assess if evidence of learning by the RL algorithm is due to random chance. For each of the 72 participants with viable data from the Oralytics clinical trial, we use that participant's data to create a participant-environment model. We then re-simulate the Oralytics trial by generating participant states, the RL algorithm selecting actions for these 72 participants given their states, the participant-environment model generating health outcomes / rewards in response, and the RL algorithm updating using state, action, and reward data generated during simulation. To make the environment more realistic, we also replicate each participant being recruited incrementally and entering the trial by their real start date in the Oralytics trial and simulate update times on the same dates as when the RL algorithm updated in the real trial (i.e., weekly on Sundays).

B.1 Participant-Environment Model

In this section, we describe how we constructed the participant-environment models for each of the $N = 72$ participants in the Oralytics trial using that participant's data. Each participant-environment model has the following components:

- Outcome Generating Function (i.e., OSCB $Q_{i,t}$ in seconds given state $S_{i,t}$ and action $A_{i,t}$)
- App Engagement Behavior (i.e., the probability of the participant opening their app on any given day)

Environment State Features The features used in the state space for each environment are a superset of the algorithm state features $f(S_{i,t})$ (Appendix A.1). $g(S_{i,t}) \in \mathbb{R}^7$ denotes the super-set of features used in the environment model.

The features are:

1. Time of Day (Morning/Evening) $\in \{0, 1\}$
2. $\bar{B}$: Exponential Average of OSCB Over Past 7 Days (Normalized) $\in [-1, 1]$
3. $\bar{A}$: Exponential Average of Prompts Sent Over Past 7 Days (Normalized) $\in [-1, 1]$
4. Prior Day App Engagement $\in \{0, 1\}$
5. Day of Week (Weekend / Weekday) $\in \{0, 1\}$
6. Days Since Participant Started the Trial (Normalized) $\in [-1, 1]$
7. Intercept Term $= 1$

Feature 5 is 0 for weekdays and 1 for weekends. Feature 6 refers to how many days the participant has been in the Oralytics trial (i.e., between 1 and 70) normalized to be between -1 and 1.

Outcome Generating Function The outcome generating function is a function that generates OSCB $Q_{i,t}$ in seconds given current state $S_{i,t}$ and action $A_{i,t}$. We use a zero-inflated Poisson to model each participant's outcome generating process because of the zero-inflated nature of OSCB found in previous data sets and data collected in the Oralytics trial. Each participant's outcome generating function is:

$$
\begin{gathered}
Z \sim \text{Bernoulli}\Big(1 - \text{sigmoid}\big(g(S_{i,t})^\top w_{i,b} - A_{i,t} \cdot \max\left[\Delta_{i,B}^\top g(S_{i,t}), 0\right]\big)\Big) \\
S \sim \text{Poisson}\big(\exp\left(g(S_{i,t})^\top w_{i,p} + A_{i,t} \cdot \max\left[\Delta_{i,N}^\top g(S_{i,t}), 0\right]\right)\big) \\
Q_{i,t} = ZS
\end{gathered}
\tag{6}
$$

where $g(S_{i,t})^\top w_{i,b}, g(S_{i,t})^\top w_{i,p}$ are called baseline (aka when $A_{i,t} = 0$) models with $w_{i,b}, w_{i,p}$ as participant-specific baseline weight vectors, $\max\left[\Delta_{i,B}^\top g(S_{i,t}), 0\right], \max\left[\Delta_{i,N}^\top g(S_{i,t}), 0\right]$ are called advantage models, with $\Delta_{i,B}, \Delta_{i,N}$ as participant-specific advantage (or treatment effect) weight vectors. $g(S_{i,t})$ is described in Appendix B.1, and $\text{sigmoid}(x) = \frac{1}{1+e^{-x}}$.

The outcome generating function can be interpreted in two components: (1) the Bernoulli outcome Z models the participant's intent to brush given state $S_{i,t}$ and action $A_{i,t}$ and (2) the Poisson outcome S models the participant's OSCB value in seconds when they intend to brush, given state $S_{i,t}$ and action $A_{i,t}$. Notice that the models for Z and S currently require the advantage/treatment effect of OSCB $Q_{i,t}$ to be non-negative. Otherwise, sending an engagement prompt would yield a lower OSCB value (i.e., models participant brushing worse) than not sending one, which was deemed nonsensical in this mHealth setting.

Weights $w_{i,b}, w_{i,p}, \Delta_{i,B}, \Delta_{i,N}$ for each participant's outcome generating function are fit that participant's state, action, and OSCB data from the Oralytics trial. We fit the function using MAP with priors $w_{i,b}, w_{i,p}, \Delta_{i,B}, \Delta_{i,N} \sim \mathcal{N}(0, I)$ as a form of regularization because we have sparse data for each participant. Finalized weight values were chosen by running random restarts and selecting the weights with the highest log posterior density. See Appendix B.2 for metrics calculated to verify the quality of each participant's outcome generating function.

App Engagement Behavior We simulate participant app engagement behavior using that participant's app opening data from the Oralytics trial. Recall that app engagement behavior is used in the state for both the environment and the algorithm. More specifically, we define app engagement as the participant opening their app and the app is in focus and not in the background. Using this app opening data, we calculate p_i^{app}, the proportion of days that the participant opened the app during the Oralytics trial (i.e., number of days the participant opened the app in focus divided by 70, the total number of days a participant is in the trial for). During simulation, at the end of each day, we sample from a Bernoulli distribution with probability p_i^{app} for every participant i currently in the simulated trial.

B.2 Assessing the Quality of the Outcome Generating Functions

Our goal is to have the simulation environment replicate outcomes (i.e., OSCB) as close to the real Oralytics trial data as possible. To verify this, we compute various metrics (defined in the following section) comparing how close the outcome data generated by the simulation environment is to the data observed in the real trial . Table 6 shows this comparison on various outcome metrics. Table 7 shows various error values of simulated OSCB with OSCB observed in the trial. For both tables, we report the average and standard errors of the metric across the 500 Monte Carlo simulations and compare with the value of the metric for the Oralytics trial data. Figure 6 shows comparisons of outcome metrics across trial participants.

Notation $\mathbb{I}\{\cdot\}$ denotes the indicator function. Let $\widehat{\text{Var}}(\{X_k\}_{k=1}^K)$ represent the empirical variance of $X_1, ..., X_K$.

Metric Definitions and Formulas Recall that $N = 72$ is the number of participants and $T = 140$ is the total number of decision times that the participant produces data for in the trial. We consider the following metrics and compare the metric on the real data with data generated by the simulation environment.

1. Proportion of Decision Times with OCSB = 0:

$$\frac{\sum_{i=1}^N \sum_{t=1}^T \mathbb{I}\{Q_{i,t} = 0\}}{N \times T} \tag{7}$$

2. Average of Average Non-zero Participant OSCB:

$$\frac{1}{N} \sum_{i=1}^N \bar{Q}_i^{\text{non-zero}} \tag{8}$$

where

$$\bar{Q}_i^{\text{non-zero}} = \frac{\sum_{t=1}^T Q_{i,t} \cdot \mathbb{I}\{Q_{i,t} > 0\}}{\sum_{t=1}^T \mathbb{I}\{Q_{i,t} > 0\}}$$

3. Average Non-zero OSCB in Trial:

$$\frac{1}{\sum_{i=1}^N \sum_{t=1}^T \mathbb{I}\{Q_{i,t} > 0\}} \sum_{i=1}^N \sum_{t=1}^T Q_{i,t} \cdot \mathbb{I}\{Q_{i,t} > 0\} \tag{9}$$

4. Variance of Average Non-zero Participant OSCB:

$$\widehat{\text{Var}}(\{\bar{Q}_i^{\text{non-zero}}\}_{i=1}^N) \tag{10}$$

where

$$\bar{Q}_i^{\text{non-zero}} = \frac{\sum_{t=1}^T Q_{i,t} \cdot \mathbb{I}\{Q_{i,t} > 0\}}{\sum_{t=1}^T \mathbb{I}\{Q_{i,t} > 0\}}$$

5. Variance of Non-zero OSCB in Trial:

$$\widehat{\text{Var}}(\{Q_{i,t} : Q_{i,t} > 0\}_{i=1,t=1}^{N,T}) \tag{11}$$

6. Variance of Average Participant OCSB:

$$\widehat{\text{Var}}(\{\bar{Q}_i\}_{i=1}^N) \tag{12}$$

where $\bar{Q}_i = \sum_{t=1}^T Q_{i,t}$ is the average OSCB for participant i

7. Average of Variances of Participant OSCB:

$$\frac{1}{N} \sum_{i=1}^N \widehat{\text{Var}}(\{Q_{i,t}\}_{t=1}^T) \tag{13}$$

We also compute the following error metrics. We use $\hat{Q}_{i,t}$ to denote the simulated OSCB and $Q_{i,t}$ to denote the corresponding OSCB value from the Oralytics trial data.

1. Mean Squared Error:

$$\frac{1}{N \times T} \sum_{i=1}^N \sum_{t=1}^T (\hat{Q}_{i,t} - Q_{i,t})^2 \tag{14}$$

2. Root Mean Squared Error:

$$\sqrt{\frac{1}{N \times T} \sum_{i=1}^{N} \sum_{t=1}^{T} (\hat{Q}_{i,t} - Q_{i,t})^2} \tag{15}$$

3. Mean Absolute Error:

$$\frac{1}{N \times T} \sum_{i=1}^{N} \sum_{t=1}^{T} |\hat{Q}_{i,t} - Q_{i,t}| \tag{16}$$

Outcome Metric	Simulation Environment	Oralytics Trial Data
Proportion of Decision Times With OSCB = 0 (Equation 7)	0.473 (0.0002)	0.477
Average Non-Zero OSCB in Trial (Equation 8)	131.196 (0.018)	131.487
Average of Average Non-Zero Participant OSCB (Equation 9)	126.894 (0.043)	127.104
Variance of Non-Zero OSCB in Trial (Equation 10)	1790.723 (3.208)	1777.210
Variance of Average Non-Zero Participant OSCB (Equation 11)	834.028 (7.434)	796.132
Variance of Average Participant OSCB (Equation 12)	69.166 (0.024)	68.827
Average of Variances of Participant OSCB (Equation 13)	3865.696 (2.723)	3883.210

Table 6: Outcome metrics for data from generated from the simulation environment vs. Oralytics trial data. Each outcome metric value under the "Simulation Environment" column is computed for each of the 500 Monte Carlo simulated repetitions. We report the mean (rounded to nearest 3 decimal places) and the standard errors (in parentheses) of these metrics across the repetitions.

Error Metric	Value
Mean Squared Error (Equation 14)	6165.169 (4.485)
Root Mean Squared Error (Equation 15)	78.516 (0.029)
Mean Absolute Error (Equation 16)	48.027 (0.023)

Table 7: Simulation Environment Error Values. Error values are computed using the simulated OSCB and the OSCB values in the Oralytics trial data. An error value is computed for each of the 500 Monte Carlo repetitions. We report the mean (rounded to nearest 3 decimal places) and the standard errors (in parentheses) across these repetitions.

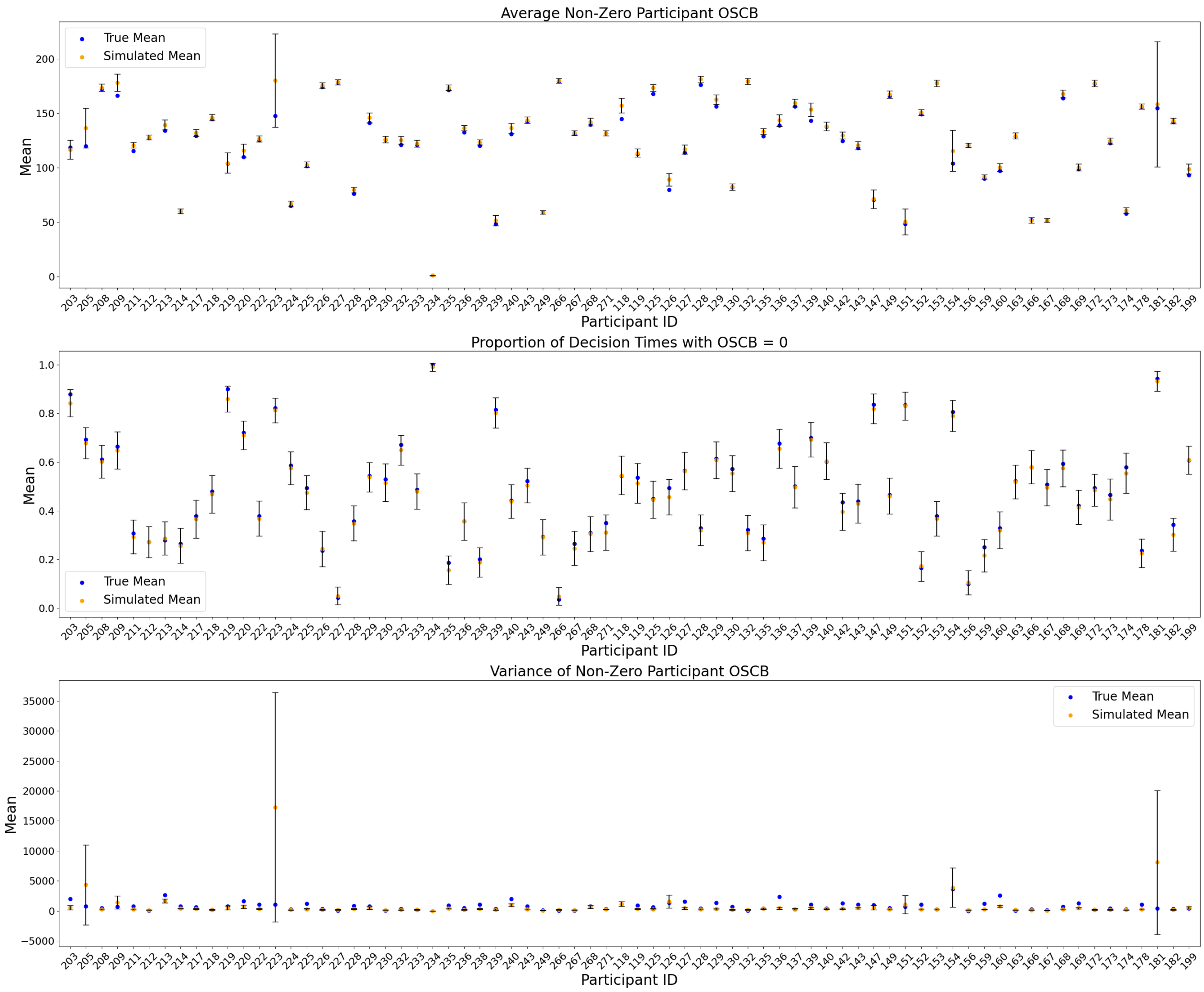

Figure 6: Outcome metrics across trial participants comparing data generated by the simulation environment with Oralytics trial data. Error bars depict confidence intervals across 500 Monte Carlo repetitions.

B.3 Environment Variants for Re-sampling Method

In this section, we discuss how we formed variants of the simulation environment used in the re-sampling method from Section 5.4. We create a variant for every state s of interest corresponding to algorithm advantage features $f(s)$ and environment advantage features $g(s)$. In each variant, outcomes (i.e., OSCB $Q_{i,t}$) and therefore rewards, are generated so that there is *no advantage* of action 1 over action 0 in the particular state s.

To do this, recall that we fit an outcome generating function (Equation 6) for each of the $N = 72$ participants in the trial. Each participant i's outcome generating function has advantage weight vectors $\Delta_{i,B}, \Delta_{i,N}$ that interact with the environment advantage state features $g(s)$. Instead of using $\Delta_{i,B}, \Delta_{i,N}$ fit using that participant's trial data, we instead use projections proj $\Delta_{i,B}$, proj $\Delta_{i,N}$ of $\Delta_{i,B}, \Delta_{i,N}$ that have two key properties:

1. for the current state of interest s, on average they generate treatment effect values that are 0 in state s with algorithm state features $f(s)$ (on average across all feature values for features in $g(s)$ that are not in $f(s)$)
2. for other states $s' \neq s$, they generate treatment effect values $g(s')^\top$proj $\Delta_{i,B}, g(s')^\top$proj $\Delta_{i,N}$ close to the treatment effect values using the original advantage weight vectors $g(s')^\top \Delta_{i,B}, g(s')^\top \Delta_{i,N}$

To find proj $\Delta_{i,B}$, proj $\Delta_{i,N}$ that achieve both properties, we use the SciPy optimize API[4] to minimize the following constrained optimization problem:

[4]Documentation here: https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.minimize.html

$$\min_{\text{proj }\Delta} \frac{1}{K}\sum_{k=1}^{K}(g(s')_k^\top \text{proj }\Delta - g(s')_k^\top \Delta)^2$$

$$\text{subject to: } \tilde{g}(s)^\top \text{proj }\Delta = 0$$

$\{g(s')_k\}_{k=1}^K$ denotes a set of states we constructed that represents a grid of values that $g(s')$ could take. $\tilde{g}(s)$ has the same state feature values as $g(s)$ except the "Day of Week" and "Days Since Participant Started the Trial (Normalized)" features are replaced with fixed mean values $2/7$ and 0. The objective function is to achieve property 2 and the constraint is to achieve property 1.

We ran the constrained optimization with $\Delta = \Delta_{i,B}$ and $\Delta_{i,N}$ to get proj $\Delta_{i,B}$, proj $\Delta_{i,N}$, for all participants i. All participants in this variant of the simulation environment produce OSCB $Q_{i,t}$ given state $S_{i,t}$ and $A_{i,t}$ using Equation 6 with $\Delta_{i,B}, \Delta_{i,N}$ replaced by proj $\Delta_{i,B}$, proj $\Delta_{i,N}$.

C Additional Did We Learn? Plots

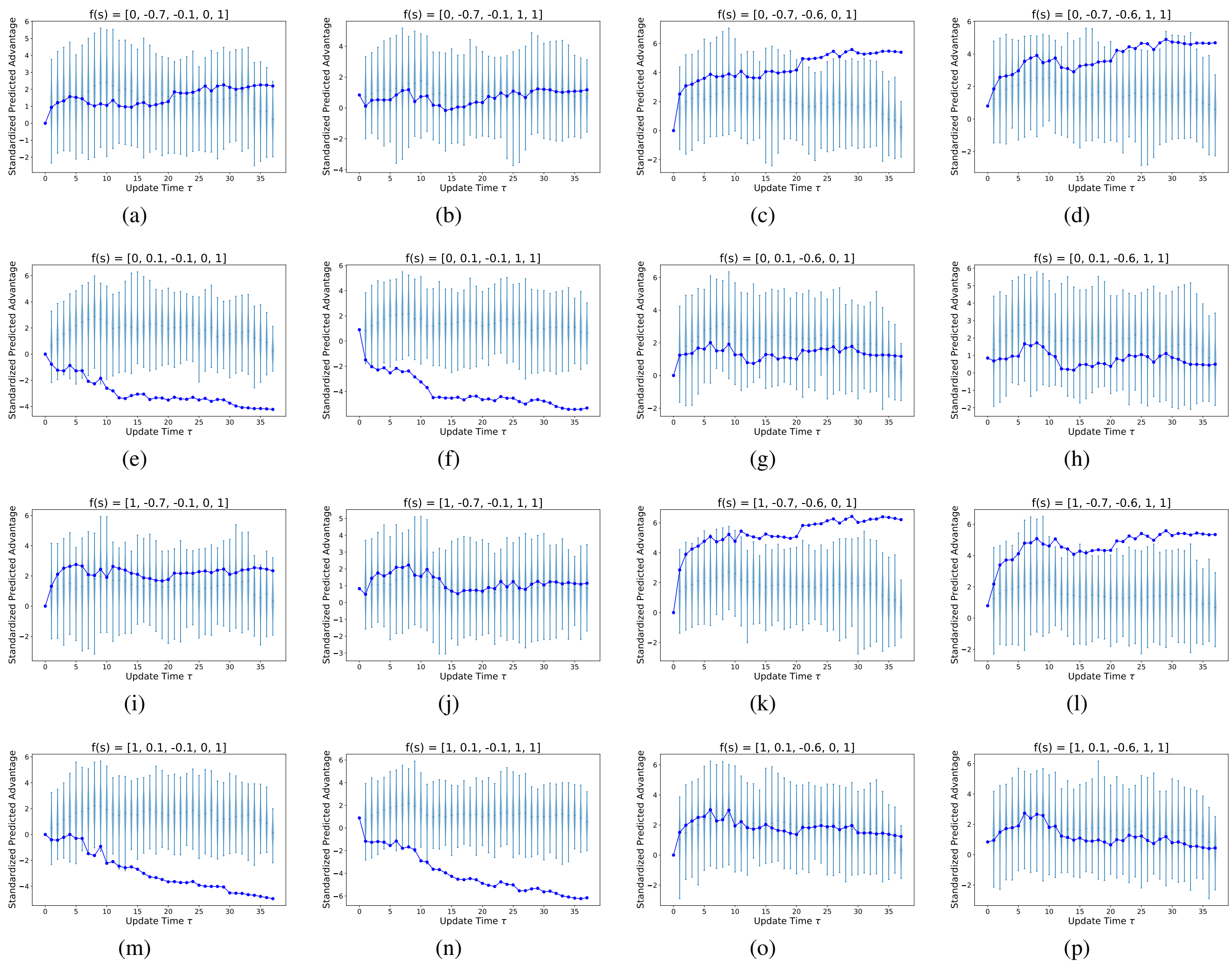

Figure 7: "Did We Learn?" using the re-sampling based method on 16 different states of interest. We compare standardized predicted advantages across updates to the posterior parameters from the actual Oralytics trial (dark blue) with violin plots of simulated predictive advantages using posterior parameters re-sampled across 500 Monte Carlo repetitions (light blue).

In Section 5.4 we considered a total of 16 different states of interest. Results for all 16 states are in Figure 7. Recall each state is a unique combination of the following algorithm advantage feature values:

1. Time of Day: $\{0, 1\}$ (Morning and Evening)
2. Exponential Average of OSCB Over Past Week (Normalized): $\{-0.7, 0.1\}$ (first and third quartile in Oralytics trial data)
3. Exponential Average of Prompts Sent Over Past Week (Normalized): $\{-0.6, -0.1\}$ (first and third quartile in Oralytics trial data)
4. Prior Day App Engagement: $\{0, 1\}$ (Did Not Open App and Opened App)

Notice that since features (2) and (3) are normalized, for feature (2) the quartile value of -0.7 means the participant's exponential average OSCB in the past week is about 28 seconds and similarly 0.1 means its about 100 seconds. For feature (3), the quartile value of -0.6 means the participant received prompts 20% of the time in the past week and similarly -0.1 means it's 45% of the time.